

| age     | income | student | credit_rating | buys_computer |
|---------|--------|---------|---------------|---------------|
| <=30    | high   | no      | fair          | no            |
| <=30    | high   | no      | excellent     | no            |
| 31...40 | high   | no      | fair          | yes           |
| >40     | medium | no      | fair          | yes           |
| >40     | low    | yes     | fair          | yes           |
| >40     | low    | yes     | excellent     | no            |
| 31...40 | low    | yes     | excellent     | yes           |
| <=30    | medium | no      | fair          | no            |
| <=30    | low    | yes     | fair          | yes           |
| >40     | medium | yes     | fair          | yes           |
| <=30    | medium | yes     | excellent     | yes           |
| 31...40 | medium | no      | excellent     | yes           |
| 31...40 | high   | yes     | fair          | yes           |
| >40     | medium | no      | excellent     | no            |

Root node

$$\text{Info}(D) = I(9,5) = -\frac{9}{14} \log\left(\frac{9}{14}\right) - \frac{5}{14} \log\left(\frac{5}{14}\right) = 0.940$$

$$\text{Info}_{\text{age}}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{3}{14} I(3,2) = 0.694$$

$$\begin{aligned} \text{Info}_{\text{income}}(D) &= \frac{4}{15} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2) \\ &= \frac{4}{15} \left( -\frac{3}{4} \log\left(\frac{3}{4}\right) - \frac{1}{4} \log\left(\frac{1}{4}\right) \right) + \frac{6}{14} \left( -\frac{4}{6} \log\left(\frac{4}{6}\right) - \frac{2}{6} \log\left(\frac{2}{6}\right) \right) + \frac{4}{14} \left( -\frac{2}{4} \log\left(\frac{2}{4}\right) - \frac{2}{4} \log\left(\frac{2}{4}\right) \right) \\ &= 0.236 + 0.394 + 0.232 = 0.862 \end{aligned}$$

$$\begin{aligned} \text{Info}_{\text{student}}(D) &= \frac{7}{14} I(6,1) + \frac{7}{14} I(3,4) = \frac{7}{14} \left( -\frac{6}{7} \log\left(\frac{6}{7}\right) - \frac{1}{7} \log\left(\frac{1}{7}\right) \right) + \frac{7}{14} \left( -\frac{3}{7} \log\left(\frac{3}{7}\right) - \frac{4}{7} \log\left(\frac{4}{7}\right) \right) \\ &= \frac{7}{14} (0.592) + \frac{7}{14} (0.915) = 0.296 + 0.493 = 0.789 \end{aligned}$$

$$\begin{aligned} \text{Info}_{\text{credit}}(D) &= \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3) = \frac{8}{14} \left( -\frac{6}{8} \log\left(\frac{6}{8}\right) - \frac{2}{8} \log\left(\frac{2}{8}\right) \right) + \frac{6}{14} \left( -\frac{3}{6} \log\left(\frac{3}{6}\right) - \frac{3}{6} \log\left(\frac{3}{6}\right) \right) \\ &= \frac{8}{14} (0.918) + \frac{6}{14} (1.00) = 0.463 + 0.492 = 0.955 \end{aligned}$$

$$\text{Gain}(\text{age}) = 0.940 - 0.694 = 0.246$$

$$\text{Gain}(\text{income}) = 0.940 - 0.862 = 0.078$$

$$\text{Gain}(\text{student}) = 0.940 - 0.789 = 0.151$$

$$\text{Gain}(\text{credit}) = 0.940 - 0.955 = -0.015$$

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